

Álgebra Relacional, Cálculo Relacional y SQL

Esquema de relaciones

- Sailors(sid, sname, rating, age)
- Boats(bid, bname, color)
- Reserves(sid, bid, day)

1. 1. Selección, proyección y unión

1.1. Colores de botes reservados por Albert

Álgebra relacional

$$\pi_{color}(\sigma_{sname='Albert'}(Sailors) \bowtie Reserves \bowtie Boats)$$

SQL

```
SELECT DISTINCT b.color
FROM Sailors s
JOIN Reserves r ON s.sid = r.sid
JOIN Boats b ON r.bid = b.bid
WHERE s.sname = 'Albert';
```

1.2. sid con rating ≥ 8 o bote 103

Álgebra relacional

$$\pi_{sid}(\sigma_{rating \geq 8}(Sailors)) \cup \pi_{sid}(\sigma_{bid=103}(Reserves))$$

SQL

```
SELECT sid FROM Sailors WHERE rating >= 8
UNION
SELECT sid FROM Reserves WHERE bid = 103;
```

2. 2. Diferencia y negación

2.1. Marineros que NO reservaron bote rojo

Álgebra relacional

$$\pi_{sname}(Sailors) - \pi_{sname}(Sailors \bowtie Reserves \bowtie \sigma_{color='red'}(Boats))$$

SQL

```
SELECT s.sname
FROM Sailors s
WHERE NOT EXISTS (
    SELECT 1
    FROM Reserves r
    JOIN Boats b ON b.bid = r.bid
    WHERE r.sid = s.sid AND b.color = 'red'
);
```

2.2. sid edad > 20 sin bote rojo

SQL (LEFT JOIN)

```
SELECT s.sid
FROM Sailors s
LEFT JOIN Reserves r ON s.sid = r.sid
LEFT JOIN Boats b ON r.bid = b.bid AND b.color = 'red'
WHERE s.age > 20
GROUP BY s.sid
HAVING COUNT(b.bid) = 0;
```

2.3. EXCEPT vs LEFT JOIN

EXCEPT

```
SELECT s.sid
FROM Sailors s
WHERE s.age > 20
EXCEPT
SELECT s.sid
FROM Sailors s
JOIN Reserves r ON s.sid = r.sid
JOIN Boats b ON b.bid = r.bid
WHERE b.color = 'red';
```

LEFT JOIN

```

SELECT s.sid
FROM Sailors s
LEFT JOIN Reserves r ON s.sid = r.sid
LEFT JOIN Boats b ON r.bid = b.bid AND b.color = 'red'
WHERE s.age > 20
GROUP BY s.sid
HAVING COUNT(b.bid) = 0;

```

3. 3. Autoreferencia y renombrado

3.1. Marineros con al menos dos botes distintos

SQL

```

SELECT DISTINCT r1.sid
FROM Reserves r1
JOIN Reserves r2
  ON r1.sid = r2.sid
 AND r1.bid <> r2.bid;

```

3.2. Rating mayor que algún Bob

SQL

```

SELECT DISTINCT s.sid
FROM Sailors s
WHERE s.rating > ANY (
  SELECT rating FROM Sailors WHERE sname = 'Bob'
);

```

3.3. Rating mayor que todos los Bob

SQL

```

SELECT s.sid
FROM Sailors s
WHERE s.rating > ALL (
  SELECT rating FROM Sailors WHERE sname = 'Bob'
);

```

4. 4. División

4.1. Marineros que reservaron todos los botes

SQL (NOT EXISTS)

```

SELECT s.sid
FROM Sailors s
WHERE NOT EXISTS (
    SELECT 1
    FROM Boats b
    WHERE NOT EXISTS (
        SELECT 1
        FROM Reserves r
        WHERE r.sid = s.sid AND r.bid = b.bid
    )
);

```

SQL (GROUP BY)

```

SELECT r.sid
FROM Reserves r
GROUP BY r.sid
HAVING COUNT(DISTINCT r.bid) = (SELECT COUNT(*) FROM Boats);

```

4.2. Botes llamados BigBoat

```

SELECT s.sid
FROM Sailors s
WHERE NOT EXISTS (
    SELECT 1
    FROM Boats b
    WHERE b.bname = 'BigBoat'
    AND NOT EXISTS (
        SELECT 1
        FROM Reserves r
        WHERE r.sid = s.sid AND r.bid = b.bid
    )
);

```

4.3. Botes de marineros con menor rating

```

SELECT s.sid
FROM Sailors s
WHERE NOT EXISTS (
    SELECT 1
    FROM Reserves r1
    JOIN Sailors s2 ON r1.sid = s2.sid
    WHERE s2.rating < s.rating
    AND NOT EXISTS (
        SELECT 1

```

```
        FROM Reserves r2
        WHERE r2.sid = s.sid AND r2.bid = r1.bid
    )
);
```

5. 5. Máximo y mínimo sin agregados

5.1. Rating máximo

```
SELECT s1.sid
FROM Sailors s1
WHERE NOT EXISTS (
    SELECT 1
    FROM Sailors s2
    WHERE s2.rating > s1.rating
);
```

5.2. Marinero más viejo

```
SELECT s1.sname, s1.age
FROM Sailors s1
WHERE NOT EXISTS (
    SELECT 1
    FROM Sailors s2
    WHERE s2.age > s1.age
);
```